

Project Plan

Introduction & Motivation:

Hotwire cutting is an automated manufacturing process where a heated wire usually made of nichrome is used to cut simple profiles out of a suitable parent material (Usually an expanded or extruded polymer product). Cutting is accomplished by heating the material to the point that the material thermally decomposes into gaseous products which naturally evacuate the cutting region, it follows that the size of the cut is proportional to the heat delivered at a point along the cut, a greater heat flux or a slower cut will increase the total thermal energy delivered to region of the parent material and so will result in a greater kerf (width of cut). Additional contributing factors result from high curvature geometries and thin geometries. The result is that when computing the tool path assuming a fixed kerf width results in distorted geometries.

By modelling the shape to be manufactured as a ruled surface defined by sets of parametrically defined b-splines the optimisation problems dimensionality can be reduced. Validation will be accomplished by comparing the measured geometry of test articles with the ground truth dimensions. In addition to testing the final product various other methods will be considered to show the relative optimisation performance.

Hotwire cutters are frequently used as a method for rapid manufacture and prototyping of UAV's (Adawy, 2023) and other foam products, current software cannot correct for deviations in the manufactured shape from the designed shape. In many industrial uses these deviations are acceptable due to the scale or simple geometry of the manufactured product. Similarly for much research scales are large and so the dynamics can be safely ignored, further when reviewing past literature little is presented of how researchers have validated that the manufactured geometries match the designed geometries. In some research error data is presented, Duenser (2023) presented a quantitative examination of cutting errors, their use of a hot blade and low cutting speeds of 2mm/s allowed for surface geometries within 2% of the designed geometry, although this research is not directly transferable it shows that low errors are possible and presents useful metrics when approaching this problem and a potential target.

Percentage error, using rudimentary shapes with known cutting characteristics, or rely on more expensive continuous wire cutters and hot blade cutters, these place solutions outside of the accessible range for many people or companies. In recent years many UAV designers have shifted from the use of hotwires for prototyping to 3D printing.

Aim:

Develop a method to optimise hotwire trajectory to minimise deviation from the designed shape.

Objectives:

OB1. Understand sources of shape errors, quantify and characterise them.

OB2. Design a model to predict errors in shape.

OB3. Research, develop, and implement a method for optimising wire trajectory.

OB4. Quantify differences in performance between the 3 correction methods, How well each method reduces error, how the errors effects foil performance.

OB5. Quantify the performance of the new algorithm.

EDIA considerations

Given the social impact of this topic is unlikely to be significant as a result of findings associated with its results and execution consideration focus primarily on accessibility.

Consideration for font spacings, font, will be made for those with reading difficulties. A colour palette that is more accessible for those with particular colour blindnesses will be chosen, if possible different version of the report with different colour schemes will be available to cater to those who have Tritanomaly as this colour scheme is mutually exclusive to those with Deuteranomaly and Protanomaly.

Tool like <https://www.color-blindness.com/coblis-color-blindness-simulator/> will be used to check accessibility

Project Timeline and work packages

Below is an initial project timeline, much of the timeline in second semester is absent due to the uncertainty inherent any timeline.

Work Package 0:	Participants:
Project management	Author
Objective:	Start: 28 th September End: Ongoing
Inputs:	
Supervisor input	
Previous experience	
Tasks:	
Develop Project timeline	
Consider required resources, support and upskilling necessary	
Outputs:	
Initial project timeline, initial schedule of meetings	

Work Package 1:	Participants:
Testing	Author
Objective: OB1 and OB4	Start: 28 th September End: Mid Semester 2
Inputs:	
Previous experience, Supervisor input	
Tasks:	
Design Standard Test article, - what will best demonstrate improvement - what is reasonable to be expected - is this a geometry that is within the capability of this manufacturing method	
Manufacture test article with fixed kerf width and old adaptive kerf width method	
Get measurements of surface shape from 3D print lab 3d scanner or CMM machine	
Compare measurement to ground truth	
Design wind tunnel mounts	
Test manufactured foils in subsonic project tunnel, gather data on wing performance	
Compare measured results with theoretical and ground truth results	
Outputs:	

Geometry and wind tunnel data

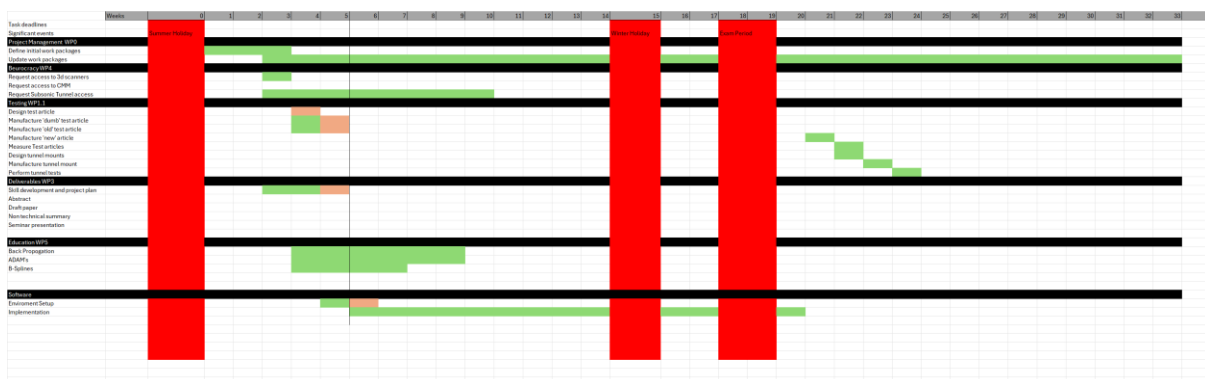
Work Package 2:	Participants:
Deliverables	Author
Objective: All of objectives	Dates: see below
Inputs:	
Supervisor input	
Tasks:	
Skill development and project plan (23 rd September - 21 st October) - A document detailing the project outline including objectives and central aim Abstract (28 th October – 23 rd November) - Details to be decided but a paper abstract fit for submission to a relevant conference Seminar Presentation (27 th January – 31 st march) - A technical presentation to a group of professionals Non-technical summary(27 th January – 5 th may) - A non-technical summary of work performed	
Outputs:	
The requested set of deliverables	

Work Package 3:	Participants:
Bureaucracy	Author
Objective: All objectives	Start: Ongoing
Inputs:	
None	
Tasks:	
Speak with ozgun and Andrew to get wind tunnel time. Speak with the 3D print lab confirm access to the 3D scanner Speak with Robert Hyneman request access to the CMM machine	
Outputs:	
Date and time for tunnel testing Process to book use of 3d Scanner	

Work Package 4:	Participants:
Research and Development	Author
Objective: OB2, OB3, and OB4	Start: September End:
Inputs:	
Supervisor input	
Problem areas identified in skill reflection	
Tasks:	
Literature review - Review relevant papers. - What is the state of the art, have others attempted similar in the past. - What methods would be suitable for this project. - How can the loss function be defined, how will it drive the optimisation Play with chosen method - Gain familiarity with the chosen method - How can the method be efficiently implemented - How will potential limits on dimensionality be addressed Implementation - Implement the chosen method integrating with old UI - Analyse the performance of the new method - How fast does the program execute - How does its performance depend on the input condition	

- Are there any anachronisms to the method, potential flaws?
Outputs:
A chosen method for approaching the project
An understanding of a method of implementation of the method
A subroutine that can perform optimisation of the chosen airfoil
Quantitative information about how the subroutine performed and scaled.

Work Package 5:	Participants:
Education	Author
Objective: OB2 and OB3	Start: 28 th September End: Ongoing
Inputs:	
Supervisor input	
Previous experience	
Tasks:	
Back propagation - Gain a theoretical understanding - Gain a practical understanding of its implementation	
ADAMS - Gain a theoretical understanding - Gain a practical understanding of its implementation	
B-Splines - Gain a theoretical understanding - Gain a practical understanding of its implementation	
Outputs:	
A working understanding of mathematical tools relevant to the project	



Reflection

There are several potential conflicts that may arise, a central part of this project is the comparative analysis, if the Wind Tunnel or 3d scanners become unavailable a central aim of the project would be jeopardised. Excluding total mechanical failure this can be avoided with early booking of the wind tunnel with a redundant plan if that is not possible, one of which would be using pressure tapping instead of direct force measurement, another option would be importing the measured geometry in to a CFD package and using a simulation.

A key piece of feedback provided was that the project objectives were unclear, have now had the opportunity to meet with my advisor I have been able to gain a more clear understanding

of the project area and approaches to this problem. Since the meeting the project objectives have been iterated.

Skill Development Plan:

Reflection:

I have significant experience in python and in use of hotwire machines for manufacturing, as part of my dissertation I gained experience with computer optimisation, this should help in keeping the development time line of a reasonable length.

I already have a large package of supporting code. Integrating new code into old code is notoriously challenging, it is possible that implementation may take far longer than expected and cause a significant over run. Having looked at the code, it is lacking in some ways and part will have to be refactored, this will need to be accounted for in the time estimates. This task is well within my abilities and should not take very long.

Skill development

Time has been allotted and specific work packages made that focus on working with the variety of mathematical tools needed for this project, hopefully with will reduce the number of false starts when developing the software. This was a problem I experienced repeatedly during my individual project and caused huge delays that resulted in the software only being finished a week or so before the final deadline. Given the dependency of testing on completion of the code this isn't acceptable. The steps taken should deal much of the problems highlighted in the reflection.

my experience with optimisation has focused on problems that are not similar to this problem, my experience lies with single DOF continuous systems, or n-DOF systems with binary design variables. This problem is described by a n-DOF system with continuous design variables . Although a very well-studied space I do not yet understand it well enough to make best use of prior work.

A persistent problem with my academic work is my weakness In academic writing, although not much can be done to remedy the root cause it can be offset by submitting more drafts for feedback, extending the writing timeline will also help me catch more mistakes in writing and to better communicate concepts.

Relevant Citations

Duenser, S., Poranne, R., Thomaszewski, B., and Coros, S., 2020. Robocut: Hot-wire cutting with robot-controlled flexible rods. *ACM Transactions on Graphics (TOG)*, 39(4), pp.98-1. Available at: <https://crl.ethz.ch/papers/hotwirecutter.pdf> (Accessed 25/10/2024)

El Adawy, M., Abdelhalim, E.H., Mahmoud, M., Mohamed, I.H., Othman, M.M., ElGamal, G.S., and ElShabasy, Y.H., 2023. Design and fabrication of a fixed-wing Unmanned Aerial Vehicle (UAV). *Ain Shams Engineering Journal*, 14(9), p.102094. Available at: <https://www.sciencedirect.com/science/article/pii/S2090447922004051> (Accessed:25/10/2024)

Figliolini, G., Rea, P., and Cocomello, C., 2018. Mechatronic design and prototype of a 4-DOFs hot-wire CNC cutting machine. In *Advances in Service and Industrial Robotics: Proceedings of the 26th International Conference on Robotics in Alpe-Adria-Danube Region, RAAD 2017* (pp. 591-598). Springer International Publishing. Available at: https://link.springer.com/chapter/10.1007/978-3-319-61276-8_62 (Accessed: 25/10/2024)

Karmakar, N., and Subbiah, S., 2018. Investigating Bowing of Hot Wire during cutting of EPS. *Procedia Manufacturing*, 26, pp.671-680. Available at: https://www.researchgate.net/publication/326822350_Investigating_Bowing_of_Hot_Wire_during_cutting_of_EPS (Accessed: 25/10/2024)

Mollica, E., and Timmoneri, A., 2021. CFD analysis of the low Reynolds S1223 airfoil. *International Journal of Engineering, Science and Technology*, 13(4), pp.46-49. (Accessed: 25/10/24)

Wang, C., 2019. MAEG5030: Geometric Computing for Design and Manufacturing. Available at: <https://mewangcl.github.io/Courses/CUHKGeoCompDM.htm> (Accessed: 25/10/24)